

Osmania University

SUPPLEMENTARY EXAMINATION ? 2025

Power Electronics ? Supplementary Examination, 2025

Subject Code: EE402 Semester: 4 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Power Semiconductor Devices)

1. Compare and contrast the key concepts involved in Power Semiconductor Devices. [8 Marks]
2. Explain Power Semiconductor Devices with a neat diagram/sketch and a suitable example. [8 Marks]
3. List the advantages and limitations of Power Semiconductor Devices. Explain any two in detail. [8 Marks]

SECTION B (Unit 2: Controlled Rectifiers)

4. Explain Controlled Rectifiers with a neat diagram/sketch and a suitable example. [12 Marks]
5. Solve a problem related to Controlled Rectifiers. Show all the steps clearly. [8 Marks]
6. Explain Controlled Rectifiers with a neat diagram/sketch and a suitable example. [8 Marks]

SECTION C (Unit 3: Choppers)

7. List the advantages and limitations of Choppers. Explain any two in detail. [12 Marks]
8. Define Choppers. Explain its significance in Power Electronics with a suitable example. [12 Marks]
9. Compare and contrast the key concepts involved in Choppers. [12 Marks]

SECTION D (Unit 4: Inverters)

10. Derive the expression related to Inverters and explain each term. [12 Marks]
11. Define Inverters. Explain its significance in Power Electronics with a suitable example. [12 Marks]
12. Describe the working principle of Inverters with necessary equations. [12 Marks]

SECTION E (Unit 5: AC Voltage Controllers)

13. List the advantages and limitations of AC Voltage Controllers. Explain any two in detail. [12 Marks]
14. Compare and contrast the key concepts involved in AC Voltage Controllers. [12 Marks]

Marks]

15. Describe the working principle of AC Voltage Controllers with necessary equations.

[8 Marks]

*** End of Question Paper ***